

ClickSense: A Low-Cost Tangible Active User Input Method Using Passive Acoustic Sensing for Mobile Virtual Reality

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Abstract

Mobile virtual reality (MVR) employs smartphone-integrated head-mounted displays to provide low-cost immersive experiences widely available to the public. Compared to higher-fidelity devices, MVR offers poor interaction support notably lacking a mechanism to produce click events necessary to indicate selection. We propose ClickSense: a no-cost active input method based on acoustic sensing. ClickSense uses the smartphone's built-in microphone to passively listen to environmental audio to identify the unique click sound emitted from a handheld clicker. Keeping with the spirit of the low-cost nature of MVR, the clicker itself is made of recycled household objects for no cost. We present a sound recognition system that can easily run on lower-powered mobile devices. Unlike vision-based gestures, our approach is not dependent on the smartphone's camera fidelity. Our evaluation of the system reveals that it offers a 97.33% recognition rate when used in isolation. When used to indicate selection in a standard selection task, it offered a 95.05% recognition.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction**; • **Interaction devices**; • **Sound-based input**;

Keywords

Virtual reality, Google Cardboard, acoustic sensing, sound input, tangible interaction

ACM Reference Format:

Kristen Grinyer and Robert J Teather. 2025. ClickSense: A Low-Cost Tangible Active User Input Method Using Passive Acoustic Sensing for Mobile Virtual Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720000>

1 Introduction

Virtual reality (VR) and other extended reality (XR) technologies are gradually gaining a foothold in everyday life; applications for beyond-entertainment, like productive work [32], personal fitness

[33], leisure, and socializing [9], are beginning to emerge. As suggested by Forbes, VR will be among the most transformative technologies of the next decade, impacting all aspects of society [26]. However, price remains a large barrier to VR adoption [16]. Most modern VR head-mounted displays (HMD) cost between \$500-1500 USD, with higher-end models selling for \$3500 USD. Mobile VR (MVR) is a low-cost alternative that uses the user's smartphone as a display and computing hardware. MVR devices range from \$15-20 USD; MVR's low cost offers the potential to make VR technologies more widely available across socio-economic status. However, MVR offers extremely limited interaction support compared to high-end devices [10, 11, 19, 25, 30, 38] and lacks any type of controller necessary for most basic interactions in VR. Some MVR devices use a conductive lever to support screen tapping, but others lack any button and provide no form of active user input.

This extended abstract builds on our past work towards improving interaction on MVR [11]. We previously proposed a 6-degrees-of-freedom (DOF) handheld controller made of cardboard and tracked by the smartphone's outward facing camera. The Cardboard Controller supports both ray-based and virtual hand selection – the most common VR selection techniques [1, 20, 25] – using dwell time as a selection indication method. Selection indication is the critical final step of a selection task; after positioning the cursor, ray, or hand on the target to be acquired, some mechanism is required so the user can indicate completion of the task [4], i.e., to “click” on the target. Modern VR controllers universally support button-based selection indication, and hand-based gestures (e.g., pinch, air tap) are increasingly common. However, the Cardboard Controller lacked any *active* selection indication mechanism. Active selection indication methods have been shown to be superior to timer-based methods (e.g., dwell), as they give users more control over where they are selecting, improving selection accuracy [11].

We propose ClickSense: a new method for active selection indication using the smartphone's microphone to detect audio made by a tangible apparatus. We made a handheld prototype button from common materials, that when pressed with the thumb, issues a unique clicking noise which is identified by the system as user input via passive acoustic sensing. ClickSense offers the ability to “click” in MVR while providing tactile feedback and meeting the low-cost objective of MVR. We informally evaluated the accuracy of the system's sound recognition when used in isolation and in a selection task. We found 97.33% and 95.05% of clicks correctly registered in the two scenarios respectively.

This extended abstract makes the following contributions:

- Insight into the use of sound for active user input for mobile VR.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3720000>

- Evaluation of a low-cost tangible button and sound recognition system.

2 Related Work

2.1 Sound Recognition

Sound has been well-established as an input method. An early example of a commercial product employing sound input is the Zenith Space Command TV remote¹ released in 1956. The remote was a mechanical device that emits ultrasonic sounds by pressing switches that in turn activated spring-loaded hammers to hit internal tuning forks. The sound issued by the tuning forks was detected by a microphone in the TV. Although certain environmental noises interfered with the remote and the ultrasonic sounds were noticeable by pets, the Zenith Space Command offered multiple buttons via a low fidelity clicking mechanism, similar to ClickSense, demonstrating that auditory input can support simple user input.

While tracked gestures must be spatially unique in three dimensions, sounds produced from gestures must be acoustically unique and are effectively one-dimensional [13]. While this limits the number of possible gesture-elicited sounds, referred to as *audio gestures* [31], it requires less computational effort and is suitable for lower-powered mobile devices.

Passive acoustic sensing has been used to recognize sounds produced by interacting with tangible objects [34, 35] and sliding and tap gestures on surfaces [7, 8, 13, 14, 23, 24, 31, 39]. Harrison and Hudson [13] proposed Scratch Input, where *audio gestures* were produced by the unique acoustic signals made from scratching a fingernail along a surface. To detect each unique acoustic signal, they used a simple decision tree based on amplitude peak count and variation. To reduce false positives, they employed several methods including noise suppression, input windowing, and environmental noise under 3KHz was ignored, achieving an 89.5% accuracy rate.

Scratch Input inspired past work, and the current work, using passive acoustic sensing to recognize user interaction [7, 14, 23, 24, 31, 35, 37, 39]. Detailed in section 4, we employ similar filters based on sound duration and amplitude patterns to detect the sound emitted by our clicker prototype and apply similar methods to reduce false positives.

2.2 Selection Indication Methods

Selection indication in VR commonly employs physical controls, sound or voice commands, vision-based gestural commands, or hybrid techniques [20]. Physical controls, like buttons or switches, provide tactile feedback and are more reliable than other methods [17], but typically require the user to hold a device. Sound input is hands-free, but environmental noise causes interference. Gestural commands, such as hand or face gestures, do not require a hand-held device, and mid-air interactions have been found more engaging and faster than handheld devices [25]. However, mid-air interactions yield fatigue [2, 8, 36], and gestural commands are less accurate and reliable than physical controls [19, 25, 29]. The imprecision of vision-based gesture detection is amplified in MVR given its smaller field of view (FOV) and limited processing power [2, 10, 11, 25].

2.3 Selection Indication in MVR

Some Google Cardboard viewers have a built-in conductive lever on the top of the viewer that can simulate tapping or holding down on the touchscreen. Past studies have found the lever unreliable [3, 25] and it can jostle the user's viewpoint when being pressed down [19]. As such, researchers have explored alternative selection indication methods for MVR.

2.3.1 HMD Attachments. Two prior studies attached a touchscreen to the back of the HMD to directly manipulate UI elements [12, 21]. In both cases, the touch screen mapped 1-1 to the camera FOV. Tapping and dragging gestures were performed on-screen to select objects and menu elements at the corresponding locations. However, neither study used a Google Cardboard, as the front-mounted touchscreen (smartphone or touchpad) was too heavy for a cardboard headset [12]. Other drawbacks included arm fatigue [12], and difficulty disambiguating objects for selection in dense virtual environments.

Other researchers considered a low-cost cardboard attachment [22] mounted on the right side of the HMD. The attachment was made from a circular cardboard track with a metal washer that slides around in the track. The smartphone's magnetometer tracked the washer's position to support bidirectional scrolling input. Tapping the centre of the track would indicate selection. When compared to head-gaze selection in a menu navigation task, this technique had a lower mental workload, was more effective for larger menus, but required more physical effort to operate.

2.3.2 Gesture-Based Approaches. Software-based approaches for selection indication have the potential to eliminate the need for additional hardware. A promising avenue is optical tracking to support hand tracking in MVR [6, 25]. De Castro et al. [6] supported 6DOF hand tracking with pinch gestures as user input. However, this required an external computer to offload computational effort from the smartphone. Luo et al. [25] tracked the user's hand and fingers to support hand- and ray-based selection. The authors compared two selection indication approaches, mid-air tap gestures and the cardboard button, but neither performed adequately. Like other work [19], gesture detection was unreliable, but mid-air gestures were more engaging and faster than using a handheld device [25]. Other studies report contradictory results [18].

Other research focused on low-cost MVR input using the HMD surfaces. Two studies explored finger tapping and sliding on the Google Cardboard. CardboardSense [38] used a set of six tap gestures; their vibrations were detected by the smartphone's motion sensors. GestOnHMD [8] used a set of 33 sliding and tap gestures detected by the smartphone's microphone. Both methods offered gesture detection rates over 98% and supported UI element selection and menu scrolling. GestOnHMD was published five years after CardboardSense and the authors noted using the phone's motion sensors has a low sampling rate and limited capacity to recognize complex gestures [8]. This is supported by their significantly larger set of gestures in comparison.

The current work was inspired by GestOnHMD [8]. We use the smartphone's microphone to detect sound caused by user interaction (e.g., pressing a button), but we moved the location of the sound source from the HMD to the hand as we found tapping on the

¹<https://www.theverge.com/23810061/zenith-space-command-remote-control-button-of-the-month>

Google Cardboard can unintentionally move the user’s viewpoint and keeping your arm held up to the HMD is fatiguing over time.

2.3.3 Vocal Input. Zielasko et al. investigated non-verbal voice input; the audio signature of blowing into a headset microphone was used as a trigger alternative [40]. Their method, BlowClick, effectively supported standard selection tasks and avoided the latency and mis-recognition issues experienced with voice commands [40]. When compared to a button trigger on a handheld device, it was more exhausting and less intuitive for some participants. While the initial study took place in a CAVE, the technique was later evaluated for smartphone environments [41].

Blowing into a headset microphone is not practical in MVR as wearing the HMD and microphone may be bulky or uncomfortable. However, BlowClick shows the low-latency advantages of non-verbal audio input and the higher intuitiveness of physical button input. ClickSense holds both advantages and can be used in VR or MVR without extra equipment needed.

2.3.4 Handheld and Wearable Devices. Past literature has evaluated novel techniques to support pointing- and grabbing-based selection in MVR; a common method is using extra hardware [2, 15, 19, 27]. However, extra hardware usually imposes additional and significant costs, undermining MVRs purpose to be broadly available across socioeconomic status.

Past research has used a smartwatch [15, 19] or second smartphone [27]; user input was performed with a screen tap [15, 27] or mapped to the user’s forearm rotation [19]. Bai et al. [2] attached a Leap Motion sensor to the front of a Gear VR headset and used a Gear VR controller with an Intel RealSense tracker and Intel Compute Stick to support 6DOF pointing. The user’s nondominant hand performed pinch gestures detected by the Leap Motion. Objects were selected by pressing the controller’s trigger and selected objects were manipulated via pinch gestures. They identified challenges with mid-air gestures including arm fatigue and inaccuracies caused by the hand moving outside the camera’s FOV. Both challenges align with findings from past research [8, 11, 36].

The Cardboard Controller [11] provided a no-cost 6DOF tracked input device for MVR with a secondary handheld marker detected via optical tracking for active selection indication. The study focused on selection indication, comparing the secondary handheld marker to dwell, revealing that the handheld marker yielded significantly worse selection performance. Moreover, tracking two handheld objects (the controller and selection indication marker) caused occlusion, increased tracking loss, and increased mental workload. Given the small FOV of MVR, it was difficult for participants to keep both pieces in camera view. We speculate that using hand gestures with the controller may have similar pitfalls. The current work directly extends this previous study by devising an auditory mechanism for indicating selection.

3 Tangible Sound Source for Selection Indication

Given the issues with hand tracking or multi-object tracking in MVR [2, 6, 11, 25], our proposed selection indication method, ClickSense, is a hybrid of physical controls and sound. It provides the benefits of a physical button, tactile feedback and reliability [17],

is not dependent on a wire or Bluetooth connection, and takes advantage of the collective familiarity and intuitiveness of button input. Detecting pre-defined sounds with the smartphone offers provenly high accuracy [8] and will have consistent accuracy across smartphone models unlike camera-tracked gestures that rely on the camera quality and FOV, which varies between smartphones. The button used with ClickSense can be integrated into a handheld Cardboard Controller [11] and thereby does not require mid-air interaction reducing the risk of fatigue [2, 8, 36].

When exploring potential sound sources suitable for active user input, we focused on generating a sound waveform that could be easily and accurately replicated by users while remaining unique and distinctive from environmental noise such as speaking. Furthermore, the sound must be made with one or no hands so the Cardboard Controller can be held.

We first considered a small handheld shaker made of a plastic vial filled with metal needles. Shaking the vial created a distinct waveform that was reproducible across multiple users. However, the arm movement required significantly jostled the HMD and user viewpoint. For the same reason, we decided not to use tap gestures on the cardboard surface despite past work having success with this method [8, 38].

We then considered the mechanics of a dog clicker, a small device typically made of a plastic button and metal. When the button is pressed, the metal is flexed and emits a loud, high-pitched click. We created a similar device using household materials to emit a lower-pitched noise, as we found a dog clicker was too loud and straining on one’s ears over repeated use. We found when a jar lid is removed from a sealed jar, its centre is flexed outward and convex. When pressing down on two opposite sides of the lid’s centre simultaneously, the lid makes a sufficiently loud popping noise. Since pressing down on two points of the lid requires two thumbs and thus two hands, we made a device using cardboard, the jar lid, and a keyring that when pressed down on with one thumb, creates the clicking noise. See Figure 1.

The jar lid and key ring were fastened to a piece of cardboard in such a way that the key ring can be pressed down onto the lid. When the ring is pressed down with the thumb, a loud clicking sound is made. As this work is a proof of concept, we used DIY materials that facilitate rapid prototyping to construct the clicker.

4 Defining Sound Recognition Filters

To identify the unique acoustic pattern produced by the clicker, we developed an application in Unity that passively records audio picked up from the smartphone’s microphone and sends characteristics of the audio related to its amplitude pattern to an external spreadsheet. This data was used to define filters for isolating and detecting the click sound. We recorded data in three phases: an initial phase used to define filter values for sound recognition, a refinement phase where these filters were further refined to increase their accuracy at identifying the click sound and ignoring ambient noise, and a third phase to gauge the recognition accuracy of the sound recognition system and ease of use of the clicker when used for selection indication in a selection task. The hardware used for all tests were a Samsung Galaxy S23 Ultra, a Google Cardboard, and the clicker prototype.



Figure 1: Clicker prototype made of cardboard, a metal lid, and key ring. Right image shows how user presses down to elicit a click.

4.1 Initial Phase

In the first data collection phase, the microphone passively listened for and recorded audio sample data. Unity audio sample data is represented by an array of floating-point values ranging from -1.0 to 1.0. While these values are arbitrary, they mimic the audio's amplitude pattern and indicate its volume level. As such, we refer to the mean value of the sample data array as the volume of that sample. We multiplied the mean sample values by 100 to account for the sensitivity of the smartphone microphone. Similar to prior work [13], to reduce false positives and filter out ambient noise, we ignored sample data with an absolute mean value below 0.02, as values below this threshold included quiet environmental noise. As the microphone was always listening, a unique sound clip began when recorded audio had a volume value above 0.02 and finished when the volume dropped below that threshold. Audio clips with a duration less than 100ms were ignored to further filter out ambient noise.

4.1.1 Data Collection. We recorded 300 sound samples (clicks) over three sessions. To record a sample, a researcher pressed on the clicker prototype using their non-dominant hand in short intervals. In each session, 100 samples were recorded and the clicker was held in a different position from the microphone. Samples were recorded in a quiet environment. For each sample, we recorded metrics based on amplitude variation, as it has been shown as effective for sound recognition [13]. This includes its duration in milliseconds, the highest volume value, and the timestamp in milliseconds of when that highest value occurred.

4.1.2 Defining Sound Filters. Of the 300 samples, there were 31 unique sound durations ranging from 100-334ms. We used this time range to define a filter to be used in the second phase of data collection. Only audio detected by the microphone with a duration within 100-334ms were registered as input.

When examining the peak volume value of each sample, there were 93 unique values. After removing any values with a single instance, 47 unique values remained ranging from 0.05-1.03. We used this range to define a second filter for the second phase of data

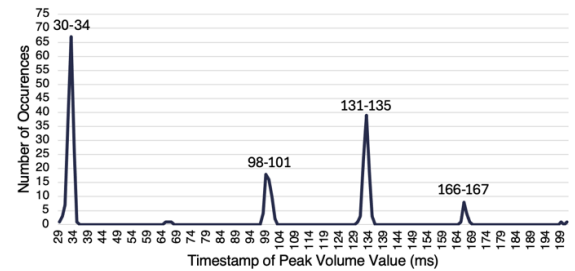


Figure 2: Timestamp of peak volume values by number of occurrences. Four clusters of values with over two occurrences are identified.

collection. Any audio recorded with a peak volume value outside of this range was ignored.

The third metric recorded was the timestamp at which the peak volume value occurred. Twenty-seven unique timestamps were recorded, with ten occurring a single time. When ignoring timestamps with two or fewer occurrences, we identified four clusters of values (see Figure 2). With these clusters, we defined a third filter. Only audio whose peak volume value occurred at a timestamp within one of the four clusters was registered as input.

4.2 Refinement Phase

In the second phase of data collection, we used an updated version of the Unity app that incorporated the filters defined in phase one. As in the first phase, the microphone passively listened for and recorded audio sample data. We recorded the same three metrics as in phase one and additionally recorded whether the sound was recognized as input (i.e., it passed all filters) or if the sound was ignored. If the latter, we recorded which filter it did not pass. We also recorded the total number of false positives caused by an environmental sound falsely registered as input.

We recorded 300 samples. Of those, 25 clicks (8.34%) were not registered as input. Eleven samples had a peak volume value out of range, where ten were too high. Another 14 samples had a peak

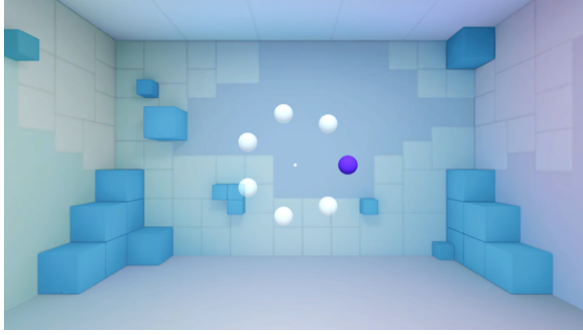


Figure 3: Selection task with eight selections. Sphere size and distance varied between tasks.

volume occurring outside the four time-range intervals identified in Figure 2. Of those, 11 occurred between 66-69ms. There were no false positives, but this was expected as we recorded the samples in a quiet environment. Based on these findings, we adjusted the filter values of these two metrics and slightly expanded them to allow for a wider range of values to be accepted as input. The updated peak volume value range is 0.05-1.25, while the updated peak volume value timestamp intervals are as follows: 30-35ms, 65-70ms, 95-105ms, 130-136ms, and 165-170ms. With these new filter values, 97.33% of the 300 samples would have correctly registered as input.

4.3 Sound Recognition in Selection Task

We performed a second test to evaluate the recognition rate of ClickSense when used as a user input apparatus and gain initial insight into its ease of use. Previous data was collected by a researcher intermediately pressing the clicker prototype. For this final evaluation, we informally tested ClickSense in a series of standard selection tasks. The virtual environment was an empty room with static spheres in front of the user (see Figure 3). The task was to select each sphere using the head-gaze technique to point at the object, and the clicker to confirm the selection. Head-gaze is the most common selection technique used in MVR [3, 8, 12, 19, 30, 38] and requires the user to point the ray originating from the centre of their head towards an object to select it. While the clicker is intended to be incorporated into the Cardboard Controller, our initial tests excluded it to minimize any confounding variables; the tracking performance of the controller could potentially affect the perceived ease of use of the clicker.

A researcher from our team performed a series of 96 selection tasks, consisting of 3×4 combinations of target distance and size, with eight individual targets in each. We did not record any selection performance metrics since the objective of this test was to assess the recognition rate of the ClickSense system. The selection tasks were completed twice.

The first time, out of 96 selections, six clicks failed to register, necessitating a second click to select the target. This yields a total sound recognition rate of 94.12%. All ignored clicks were caused by their peak volume value being too high. When repeating the 96 selections in the second round, a total of five of 101 clicks failed to register yielding a sound recognition rate of 95.05%. Again, all

clicks ignored by the system was due to a peak volume value being too high.

Despite some clicks not registering, we found using the clicker with head-gaze pointing was generally smooth and easy. Anecdotally, we found it to be more efficient than using a 300ms dwell time to confirm selections, which aligns with past work finding dwell time significantly slower than button clicks for object selection [28]. However, this informal observation will need to be formally evaluated in future work.

5 Discussion

Overall, these results are promising and demonstrate detecting user interaction via acoustic sensing is a viable option as active user input for the Cardboard Controller as ClickSense displayed a 95-97% accuracy rate. It is possible a higher recognition rate could be achieved if the filter metrics were broadened to accommodate higher peak volume values, as this was the leading cause of the click sound not registering as input. However, broadening the constraints of what sounds are classified as input also raises the risk of false positives, increasing the likelihood that ambient noise and outside voices are incorrectly identified as input.

While these initial results are promising, we identified potential limitations to this input method. Namely, the clicker cannot be operated when speaking or having significant background noise like music. In its current state, the system cannot identify the click sound if other noise over a certain volume threshold is being recorded simultaneously. This means that users cannot speak with others outside of the VR experience when operating this technique. However, this issue can be resolved with additional filters to tune out louder environmental noise.

A second potential limitation is it is unclear how microphone sensitivity varies between smartphone models and the sensitivity of the microphone directly effects how accurate the sound recognition filters will be. It is possible the filters used to isolate the click sounds will be less accurate on other smartphones, or in the extreme case, the filter metrics may need to significantly change to accommodate other microphone sensitivities.

This problem can be addressed with user-made sounds defined in real-time. The software can be modified to record a number of sound samples at the start of the experience, and define filter values in real time, to allow users to define their own sound as a method for selection indication. This would negate the smartphone model as a factor and the necessity of having a specific device to produce the sound.

6 Future Work

Our next steps include conducting a formal user study to evaluate the performance of ClickSense when used for selection indication. One avenue for future work is to compare the current sound recognition software to a version with looser constraints that allows a wider variety of sounds to be registered as input. This comparison could determine whether broader filters will positively affect the accuracy of the software and/or increase ambient sound being falsely registered as input. This study would further investigate the trade-off between higher sound recognition accuracy and false positives from ambient noise. We also plan to explore methods to

increase the sound variation to go beyond single-button interaction and potentially support multiple inputs (i.e., double click).

Another evaluation we plan to conduct is assessing the performance of the Cardboard Controller when using the clicker prototype versus dwell time to confirm selections. We would also like to compare one-handed versus two-handed operation of the controller with the clicker. In its current design, the clicker could be incorporated into the top of the controller for unimanual operation. Past work found no significant difference in throughput or movement time when triggering selections with the same or second hand [5]. We would like to investigate how ClickSense compares to these past findings.

Finally, comparing the performance of ClickSense to a standard baseline such as speech or BlowClick [40] and investigating the impact of the click sound on user experience are two other avenues worth exploring. Such studies will provide insight into ClickSense's efficacy as a user input method.

7 Conclusion

To improve the historically poor interaction support of MVR and its lack of an effective active selection indication mechanism, we proposed a low-cost method for active user input using acoustic sensing. Previous work found that cardboard controllers could only effectively support object selection if passive input (i.e., dwell time) was used [11]. Our handheld prototype makes a unique clicking sound detectable by a smartphone microphone. ClickSense avoids the drawbacks of optical tracking in MVR, does not require fatiguing mid-air interactions, provides tactile feedback, and the button-press action used is intuitive and familiar to users.

Our evaluation of ClickSense showed the system had a 97.33% accuracy rate for detecting clicks when the clicker was used in isolation, and a 95.05% accuracy rate when used with head-gaze selection. Overall, the results are a promising proof of concept for active user input for MVR. In future work, we plan to formally evaluate the performance and user experience of ClickSense with the Cardboard Controller and compare bimanual to unimanual operation.

Acknowledgments

We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) [PGSD-580042-2023].

Nous remercions le Conseil de recherches en sciences naturelles et en génie du Canada (CRSNG) de son soutien [PGSD-580042-2023].

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